Kayro AI Product Intern Assignment — Documentation

Technical Architecture, Product Decisions, and Delivery Guide

1. What We Were Asked

Build a polished appointments web application for a wellness practice with two interfaces sharing the same data: **Side A (Client App)** providing a mobile-first booking experience with upcoming/past visits and rescheduling, and **Side B (Nutritionist Dashboard)** managing visits with color-coded statuses (*Booked, Done, No-show*), today's visit filtering, manual booking, and bidirectional synchronization. The project required loading 5 specific sample records, integrating at least one concrete solution to the 25% no-show problem, and enabling review without authentication.

2. Your Approach

We treated Kayro as a calming wellness sanctuary rather than a cold administrative CRM. Technically, we implemented a **hybrid reactive data layer** (appointmentService.js) connecting to Supabase PostgreSQL when configured, while offering a zero-dependency, persistent local storage fallback so reviewers can immediately evaluate the build without manual setup. The design system utilizes warm cream backgrounds (#FBF9F4), deep forest green accents (#1E3D2F), and Playfair Display typography to convey elegance, warmth, and tranquility.

3. Decisions and Assumptions (Including How We Handled Sample Data)

- **Inconsistent Date Normalization:** Sample inputs varied across DD/MM/YYYY (24/10/2026), YYYY-MM-DD (2026-10-24), and DD-MM-YYYY (25-10-2026). We built an ingestion parser standardizing all entries to ISO YYYY-MM-DD internally for conflict checking and chronologic sorting while displaying localized dates (e.g. *Sat, 24 Oct 2026*).
- **Time Format Standardization:** Varied representations (4 pm, 11:30am, 14:00) were parsed into 24-hr HH:mm format internally and formatted into readable 12-hour AM/PM strings for display.
- **Graceful Phone Fallbacks:** Meera Iyer had no phone number; Riya Shah and Kabir Rao used disparate space conventions. We standardized 10-digit formats (98200 11111) and handled empty values gracefully with disabled WhatsApp triggers.
- **Practice Date Anchor:** Anchored the practice dashboard to 2026-10-24 matching the assignment context, enabling immediate demonstration of the 'Today Only' filter.

4. What You Added (and Why)

- **1-Tap Attendance Confirmation:** Allows clients to check in with 1 tap, giving the nutritionist 24-hour advance peace of mind instead of waiting at her desk.
- **1-Click WhatsApp Reminder Links (wa.me):** Enables the practitioner to launch personalized, pre-written WhatsApp check-in messages with zero typing.
- **Instant Calendar Sync (.ics & Google Calendar):** Pushes consultations directly to device calendars with 2-hour notifications to eliminate forgetting.
- **No-Guilt 2-Click Rescheduling:** Removes social awkwardness and immediately returns vacant slots to the calendar.

- **Reviewer Client Switcher:** A 1-click toggle allowing assessors to inspect the experience as Riya Shah, Arjun Mehta, Meera Iyer, or Kabir Rao without login.
- **Practice Analytics & No-Show Rate:** Displays real-time clinic KPIs (Active Bookings, Confirmed Count, Historical No-Show %).

5. What the AI Got Wrong (and How You Caught and Fixed It)

- **Default Template CSS Conflict:** Vite's default index.css injected #root { text-align: center; border-inline: ... }, disrupting alignments. Caught during visual inspection and replaced with a clean design token system.
- **Omitted Reschedule & Cancel Capabilities:** The starter code lacked rescheduling and cancellation logic. Built dedicated modals with conflict prevention.
- **Lexicographical Date Sorting Glitch:** Comparing mixed date strings produced incorrect chronological ordering. Standardized all comparisons to ISO YYYY-MM-DD.
- **Client Side Sync & Profile Hardcoding Bug:** The initial client interface hardcoded 4 static names and filtered by a single client, hiding any new appointments created in Side B. Resolved by deriving client profiles dynamically from Supabase and adding an 'All Practice Visits' mode so any appointment created on either side synchronizes in real time.

6. How to Run It

Local Development:

1. cd kayro-appointments && npm install
2. npm run dev — Open **http://localhost:5173** in your browser. All sample records are preloaded.

Optional Supabase PostgreSQL Connection:

Run supabase-schema.sql in the Supabase SQL editor, then populate VITE_SUPABASE_URL and VITE_SUPABASE_ANON_KEY in .env. The top indicator will display **Supabase Live**.

Production Build & Deployment:

Live Production URL: **https://kayro-appointments.vercel.app**
GitHub: **https://github.com/SK-A73/Kayro-AI-Nutritionist-Booking-App**
Configured with vercel.json for instant zero-configuration deployment on Vercel.